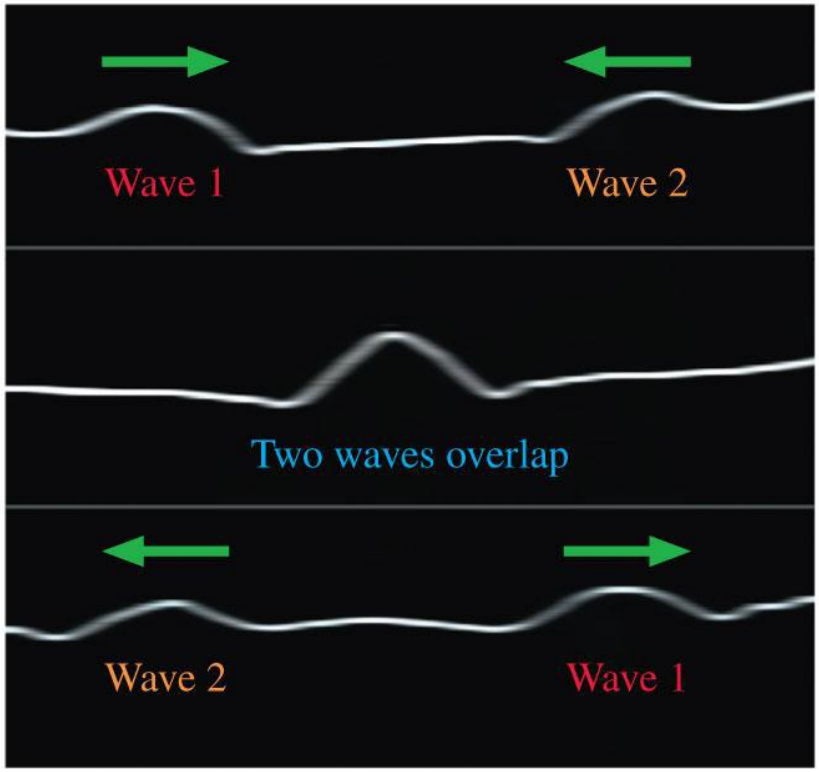
LECTURE 6

Superposition Of Waves

What happens if two or more waves pass through the same medium at the same time?



The principle of superposition

states that whenever two or more wave disturbances pass a particular point in a medium, the resultant displacement of the point of the medium is the sum of the displacements of each individual wave disturbance.

If two waves, $y_1(x,t)$ and $y_2(x,t)$, overlap, the resultant wave $y(x,t)$ is given by:

$$y(x,t) = y_1(x,t) + y_2(x,t)$$

Problem: 1

Superposition. The following two waves interfere with each other:

$$y_1 = (5.00\ \text{m})\sin[(0.800\ \text{m}^{-1})x - (6.00\ \text{rad/s})t]$$
$$y_2 = (10.00\ \text{m})\sin[(0.900\ \text{m}^{-1})x - (3.00\ \text{rad/s})t]$$

Find the resultant displacement when $x = 5.00$ m and $t = 1.10$ s.

Solution

The resultant displacement found by the superposition principle, equation 12.35, is

$$y = y_1 + y_2$$

where

$$\begin{aligned} y_1 &= (5.00\ \text{m})\sin[(0.800\ \text{m}^{-1})(5.00\ \text{m}) - (6.00\ \text{rad/s})(1.10\ \text{s})] \\ &= (5.00\ \text{m})\sin(4.00\ \text{rad} - 6.6\ \text{rad}) \\ &= (5.00\ \text{m})\sin(-2.60\ \text{rad}) \\ &= (5.00\ \text{m})(-0.5155) = -2.58\ \text{m} \end{aligned}$$

Chapter 12 Wave Motion

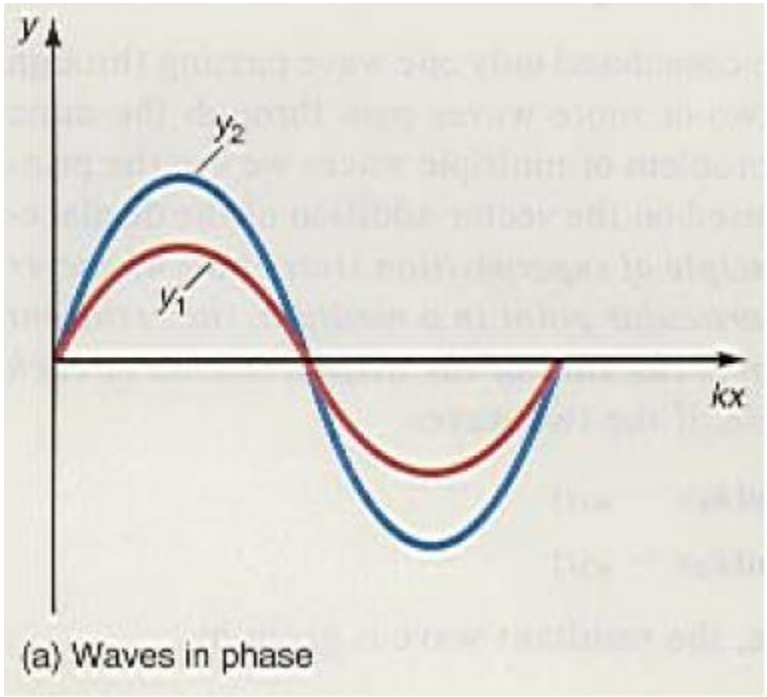
and

$$\begin{aligned} y_2 &= (10.00\ \text{m})\sin[(0.900\ \text{m}^{-1})(5.00\ \text{m}) - (3.00\ \text{rad/s})(1.10\ \text{s})] \\ &= (10.00\ \text{m})\sin(4.50\ \text{rad} - 3.30\ \text{rad}) \\ &= (10.00\ \text{m})\sin(1.20\ \text{rad}) \\ &= (10.00\ \text{m})(0.932) = 9.32\ \text{m} \end{aligned}$$

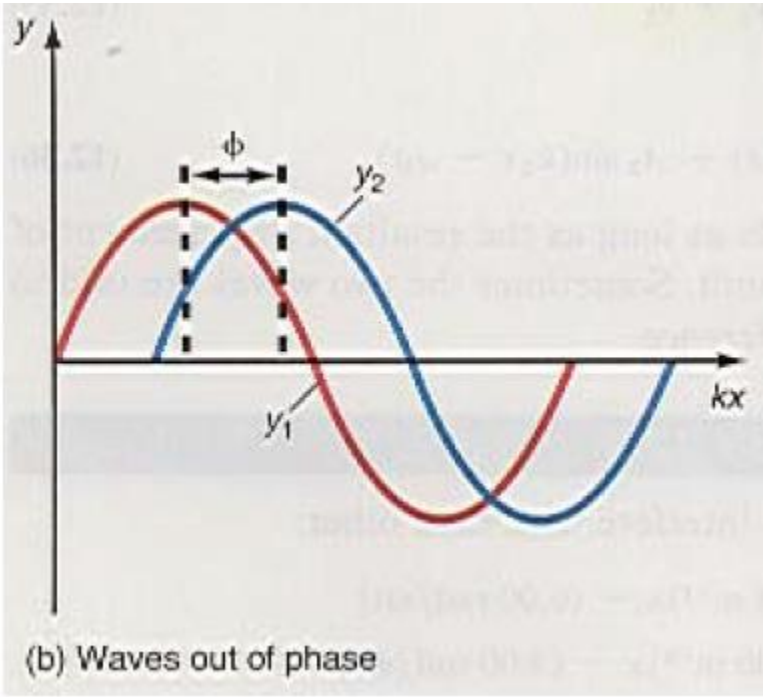
Hence, the resultant displacement is

$$\begin{aligned} y = y_1 + y_2 &= -2.58\ \text{m} + 9.32\ \text{m} \\ &= 6.74\ \text{m} \end{aligned}$$

It is possible that when dealing with two or more waves the waves may not be in phase with each other. *Two waves are in phase if they reach their maximum amplitudes at the same time, are zero at the same time, and have their minimum amplitudes at the same time.* An example of two waves in phase is shown in figure (a). An example of two waves that are out of phase with each other is shown in figure (b). Note that the second wave does not have its maximum, zero, and minimum displacements at the same place as the first wave. Instead these positions are translated to the right of their position in wave y_1. We say that wave 2 is out of phase with wave 1 by an angle ϕ, where ϕ is measured in radians.



(a) Waves in phase



(b) Waves out of phase

The equation for the first wave is

$$y_1 = A_1 \sin(kx - \omega t)$$

whereas the equation for the wave displaced to the right is

$$y_2 = A_2 \sin(kx - \omega t - \phi)$$

Phase and phase difference : Phase of the wave is given by the argument of sine or cosine in the equation of wave. It is represented by $\phi(x,t) = \frac{2\pi}{\lambda}(vt - x)$.

(v) At a given position (for fixed value of x) phase changes with time (t).

$$\frac{d\phi}{dt} = \frac{2\pi v}{\lambda} = \frac{2\pi}{T} \Rightarrow d\phi = \frac{2\pi}{T}.dt \Rightarrow \text{Phase difference} = \frac{2\pi}{T} \times \text{Time difference.}$$

(vi) At a given time (for fixed value of t) phase changes with position (x).

$$\frac{d\phi}{dx} = \frac{2\pi}{\lambda} \Rightarrow d\phi = \frac{2\pi}{\lambda} \times dx \Rightarrow \text{Phase difference} = \frac{2\pi}{\lambda} \times \text{Path difference}$$

$$\Rightarrow \text{Time difference} = \frac{T}{\lambda} \times \text{Path difference}$$

Problem 1. A transverse progressive wave on a stretched string has a velocity of 10 ms^{-1} and a frequency of 100 Hz. The phase difference between two particles of the string which are 2.5 cm apart will be

(a) $\pi/8$ (b) $\pi/4$ (c) $3\pi/8$ (d) $\pi/2$

Solution : (d) $\lambda = v/n = \frac{10}{100} = 0.1\,m = 10\,cm$

Phase difference = $\frac{2\pi}{\lambda} \times$ path difference $= \frac{2\pi}{10} \times 2.5 = \frac{\pi}{2}$

Problem 2. A simple harmonic progressive wave is represented by the equation $y = 8 \sin 2\pi (0.1x - 2t)$ where x and y are in cm and t is in seconds. At any instant the Phase difference between two particles separated by 2.0 cm in the x-direction is **[MP PMT 2000]**

(a) 18° (b) 36° (c) 54° (d) 72°

Solution : (d) $y = 8 \sin 2\pi\left(\frac{x}{10} - 2t\right)$ given by comparing with standard equation $y = a \sin 2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right)$

$\lambda = 10\,cm$

So Phase Difference = $\frac{2\pi}{\lambda} \times$ path difference $= \frac{2\pi}{10} \times 2 = \frac{2}{5} \times 180° = 72°$

Application of Superposition

- Interference
- Standing Waves
- Beats

Interference

the phenomenon where two or more waves superimpose to form a resultant wave of greater, lower, or the same amplitude.

Types of Interference

Constructive Interference

- Occurs when two waves meet in phase (crests align with crests and troughs with troughs), amplifying the resultant wave.
- **Example:** Two synchronized speakers amplify sound at certain points.

Destructive Interference

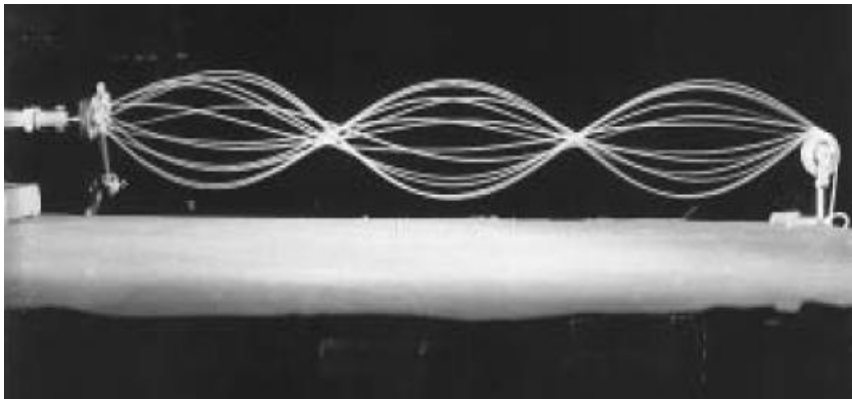
- Occurs when two waves meet out of phase (crest meets trough), canceling each other out.
- **Example:** Noise-canceling headphones use destructive interference to reduce ambient sound.

Standing Waves

Standing waves occur when two waves of the same frequency and amplitude travel in opposite directions, interfering to produce a wave pattern that appears stationary.

Characteristics of standing waves:

- Points of maximum amplitude is known as **antinodes**.
- Points of minimum amplitude is known as **nodes**.
- The amplitude of vibration of particles varies from zero at nodes to maximum at antinodes.



Beats

When two sound waves of **slightly different frequencies**, travelling in a medium along the same direction, superimpose on each other, the intensity of the resultant sound at a particular position rises and falls regularly with time. This phenomenon of regular variation in intensity of sound with time at a particular position is called **Beats**.


Thank
You